



**Baker
College**

GREATNESS
IS EARNED HERE.

Analog Principles and Operation

Programmable Logic Controllers

Brad Peirson

College of Information Technology and Engineering

Agenda

Digital vs. Analog

Analog Types

Analog Devices

PID Control

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Digital vs. Analog

Digital Defined

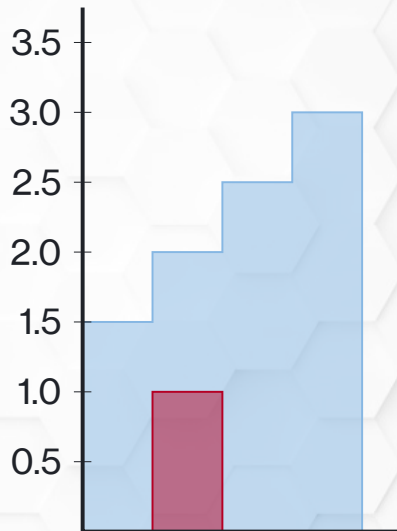
- Data or signals that have a fixed number of discrete values within their range
- Devices that use such data/signals



"Digital clock" by MichalPL, CC BY-SA 4.0, via [Wikimedia Commons](#)

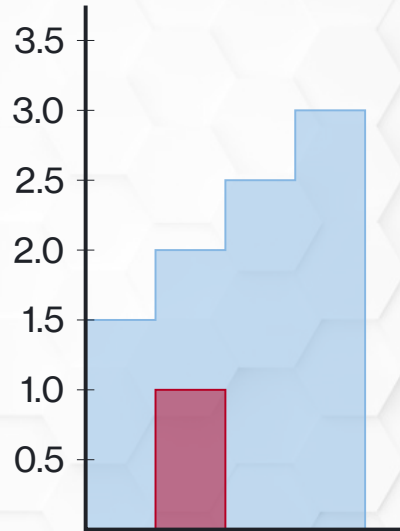
Digital Signals (1 of 2)

- *Binary* is a special case of digital—the range is 0-1
- All binary signals are digital, not all digital signals are binary



Digital Signals (2 of 2)

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



Analog Defined

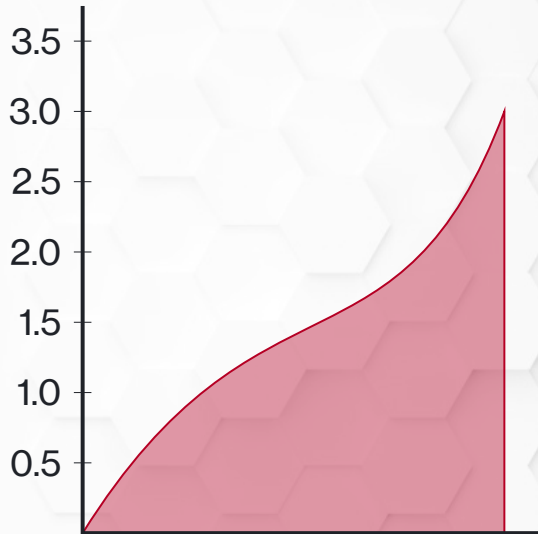
- Data or signals that vary continuously over a range of values
- Devices that use this type of data/signal



"Mercury-in-glass thermometer" by Anonimski, CC BY-SA 3.0, via [Wikimedia Commons](#)

Analog Signals

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



Why Analog?

- Motors use 69% of all electricity in the U.S. industrial sector¹
- Motors running at full speed use more energy than motors running at reduced speed

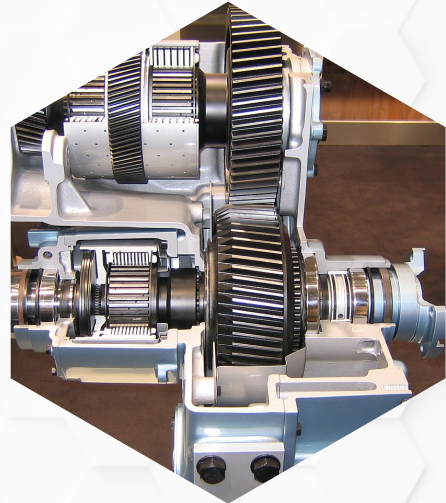


"Asynchronous electric motor" by Egzon123, CC BY-SA 3.0, via [Wikimedia Commons](#)

¹Department of Energy, "Industrial Electric Motor Market Assessment" (2020).

Controlling Motor Speed

- When a motor doesn't *need* to run at full speed you *could* use a gearbox...
- ...but that doesn't save any energy



"ZF gearbox at the Bauma trade fair 2004" by Aconcagua, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

Saving Motor Energy

- Save energy (\$) by reducing motor speed
- VFDs *always* have an analog input
- Most have 2: voltage and current
- Can accept either potentiometer or PLC analog output



"Mitsubishi frequency converter type D700" by Ignac.mihevc at Slovenian Wikipedia, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

Analog Types

Analog Signal Types

Voltage

- **0 VDC-10 VDC**
- 0 VDC-5 VDC
- -10 VDC-10 VDC

Current

- 0 mA-20 mA
- **4 mA-20 mA**

Which Are the Most Common

Which types are the most common?

Analog Example

- Two different analog sensors—identical except for their signal output
- Temperature range: 0°F-212°F
- One sensor produces a 0 VDC-10 VDC signal
- One sensor produces a 4 mA-20 mA signal

Analog Scaling (1 of 2)

- Analog devices will produce an output that directly relates to the property being measured
- Almost always linear: $1\text{ V} = 3^{\circ}\text{F}$, $2\text{ V} = 6^{\circ}\text{F}$, $3\text{ V} = 9^{\circ}\text{F}$, etc.
- To convert divide the range of the sensor by the range of the analog signal

Analog Scaling (2 of 2)

- $212^{\circ}\text{F} \div 10 \text{ VDC} = 21.2^{\circ}\text{F}/\text{V}$
- $212^{\circ}\text{F} \div 16 \text{ mA} = 13.25^{\circ}\text{F}/\text{mA}$
 - Remember, it starts at 4, the range is only 16
- Also remember that analog signals are *analog*—that means they're infinite
- These are values per unit, there an infinite number of values between

Temperature Sensor (1 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
-42.40	0	-26.5	4
-21.20	0	-13.5	4
0	0	0	4
21.2	1	13.25	5
31.8	1.5	53	8
106	5	106	12
159	7.5	159	16
201.4	9.5	198.75	19
212	10	212	20
233.2	10	225.25	20

Temperature Sensor (2 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
-42.40	0	-26.5	4
-21.20	0	-13.5	4
0	0	0	4

If the temperature decreases past the bottom of the sensor range the sensor will stay at the bottom value

Temperature Sensor (3 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
212	10	212	20
233.2	10	225.25	20

At the exact top of the sensor range the sensor will send out its maximum value

Temperature Sensor (4 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
212	10	212	20
233.2	10	225.25	20

The sensor will continue to send out its maximum value for any value above the sensor range

Temperature Sensor (5 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
-42.40	0	-26.5	4
-21.20	0	-13.5	4
0	0	0	4

Exactly *at* the bottom of the sensor range a 0 VDC-10 VDC sensor will register 0 VDC

Temperature Sensor (6 of 6)

Temp (°F)	Volts	Temp (°F)	Milliamps
-42.40	0	-26.5	4
-21.20	0	-13.5	4
0	0	0	4

Exactly *at* the bottom of the sensor range a 4 mA-20 mA sensor will register 4 mA

Which to Pick

Which of these is preferred?

Analog Signal Selection

0 VDC-10 VDC

- Easy unit conversion
- 0 is bottom, 10 is the top

4 mA-20 mA

- Current not as susceptible to noise as voltage
- Easy to diagnose—will *always* return at least 4 mA
- If you ever get a 0 something is wrong

Analog Devices



Analog Inputs

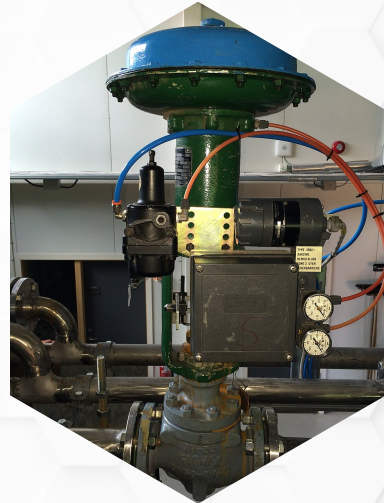
- Converts physical properties to analog signal
- A-to-D converter converts current/voltage to a binary number
- Some require external power source



"Pt-100 temperature sensor with cables" by Bitjungle, [CC BY-SA 4.0](#), via [Wikimedia Commons](#)

Analog Outputs

- Converts analog signals to physical properties
- A-to-D converter converts analog signal to current/voltage
- Some require external power source



"A Fisher control valve with positioner" by Bitjungle, [CC BY-SA 4.0](#), via [Wikimedia Commons](#)

Analog to Digital (A-to-D) Converter

- Analog *really* is infinite
- PLCs (and all computers) are digital—everything is 1's and 0's
- A-to-D converters convert the current/voltage to binary
- Each bit represents a certain number of volts/amps

Maximum Analog Value (1 of 2)

- The maximum value of the binary number from the A-to-D depends on the *bit resolution* of the converter
- Let's take the example temperature sensors and plug them into a Micrologix 1000
- The Micrologix 1000 has a 10-bit analog resolution

Maximum Analog Value (2 of 2)

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Total	1023	511	255	127	63	31	15	7	3	1

1023 is the largest number that can fit in 10 bits

Analog Resolution

- The A-to-D has a bit resolution—the number of bits it uses to store the analog conversion
- This means that each bit correlates to a specific number of volts/amps
- This *also* means that each bit correlates to a specific real-world value

Analog Resolution - Volts

- $10\text{ V} \div 1023 = 0.009775\text{ V/bit}$
- $21.2^{\circ}\text{F/V} \times 0.009775\text{ V/bit} = 0.20723^{\circ}\text{F/bit}$
- This is the *resolution*—this PLC can't see increments smaller than 0.20723°F

Bit	9	8	7	6	5	4	3	2	1	0
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098

Note: table values have been heavily rounded for formatting

Analog Resolution - mA

- $20\text{ V} \div 1023 = 0.01955\text{ mA/bit}$
- Note: this time we need the *max*—we use 20, not 16
- $13.25^{\circ}\text{F/mA} \times 0.01955\text{ mA/bit} = 0.259^{\circ}\text{F/bit}$
- This is the *resolution*—this PLC can't see increments smaller than 0.259°F

Bit	9	8	7	6	5	4	3	2	1	0
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 1 (1 of 2)

What 10-bit number represents 106°F?

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 1 (2 of 2)

- we know that 106°F is half the scale—5 V/12 mA
- Start adding up bits to get to those values and do the binary conversion
- 0-10 V sensor: 512
- 4-20 mA sensor: 614

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 2 (1 of 2)

What 10-bit number represents 72°F?

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 2 (2 of 2)

$$72^{\circ}\text{F} \div 212^{\circ}\text{F} = 0.34$$

$$0.34 \times 10 \text{ V} = 3.4 \text{ V}$$

$$3.4 \text{ V} = 342$$

$$0.34 \times 16 \text{ mA} = 5.44 \text{ mA}$$

$$5.44 \text{ mA} + 4 \text{ mA} = 9.44 \text{ mA}$$

$$9.44 \text{ mA} = 481$$

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 3 (1 of 2)

What 10-bit number represents 0°F?

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Temperature to Analog 3 (2 of 2)

For 0-10V, 0 is 0

0°F = 4 mA
4 mA = 204

Bit	9	8	7	6	5	4	3	2	1	0
Bit Value	512	256	128	64	32	16	8	4	2	1
Volts	5	2.5	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019	0.0098
mA	10	5	2.502	1.251	0.6256	0.3128	0.1564	0.0782	0.0391	0.019

Note: table values have been heavily rounded for formatting

Application

- In the real world that math almost *never* works
- These are all physical devices, no physical device is perfect
- It will get you close but usually not close enough

Analog Devices

Analog Scaling

Analog Scaling

- Analog inputs come in as integers—whole numbers that represent a voltage or current value from the sensor
- Most analog cards are 12-bit—the value received by the PLC will be 0-4096
- You *can* use this value, but it doesn't mean much to anyone else looking at the code
- It is *much* better to scale it to engineering units *before* using it in logic

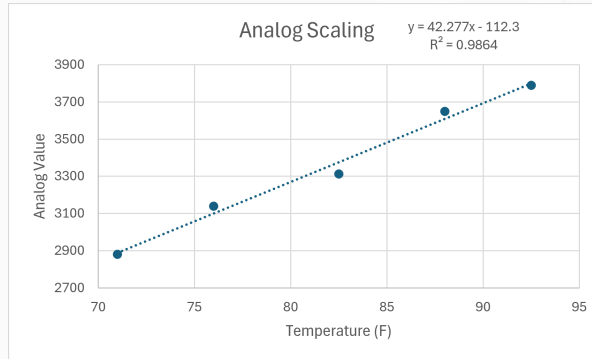
Allen-Bradley Scaling

- Allen-Bradley's preferred method in the Logix 5000 PLCs is to scale the value in the definition of the analog card itself
- Not all cards and input devices are equal—if you swap temperature sensors and it's off compared to the old one the scaling needs to be changed
- If the scaling is in the card definition, the system needs to be taken offline to change it
- Plus, not all manufacturers do it this way—we like “hardware agnostic” programming methods

Setting the Scale (1 of 2)

- Math would tell us the scaling factor would be:
$$\text{Engineering Units} = \text{UnitRange} \times \frac{\text{AnalogValue}}{\text{MaxValue}}$$
- This only works *if* the signal starts at 0
- It is also only ever going to get you close—not all sensors and cards are equal
- Best practice is to actually *measure* the conversion factor
- In this case, a spreadsheet is your best friend...

Setting the Scale (2 of 2)



Set the sensor to 5-6 known real-world values and read the analog input value. Plot them in a spreadsheet and run a *linear regression*—which the software will do automatically with one click. The regression equation is the scaling factor.

PID Control

Questions?

Copyright Notice



Analog Principles and Operation©2026 by Brad Peirson is licensed under **CC BY-NC-SA 4.0**. To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc-sa/4.0/>